

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2237

Harts Island 138 kV
December 23, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2237. The Interconnection Customer proposes a 131.5 MW solar plus storage facility interconnecting at the Harts Island 138 kV substation in the New Jersey area. The request was received on November 23, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 703 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$462,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: MAYWOOD – HARTS ISLAND 138 KV line
Contingency: COLORADO – YAHARA RIVER 138 KV
Loading: 102.5% (Rating: 289 MVA)
DFAX: 53.1%
Pre-project loading: 78.3%
2. Facility: PACIFIC POWER – MAYWOOD 138 KV
Contingency: SOUTH TACOMA – ALCHEM 138 KV line
Loading: 101.8% (Rating: 375 MVA)
DFAX: 36.6%
Pre-project loading: 89.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MAYWOOD – HARTS ISLAND 138 KV line	Reconductor line	\$603,000	58% (\$348,000)
MAYWOOD – PACIFIC POWER 138 KV	Reconductor line	\$393,000	29% (\$114,000)
	Total allocated to project		\$462,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.